

Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

<tool_call> reasoning process here

To find the de Broglie wavelength of an electron given its kinetic energy , we use the de Broglie wavelength formula :

$\lambda = h / p$

where :

- h = Planck's constant = 6.626×10^{-34} J·s (or 6.626×10^{-27} erg · Å·s),
- p = momentum of the electron .

For a non-relativistic electron , kinetic energy (K.E.) = $(1/2)mv^2$, and the momentum $p = mv$.

So , $p = \sqrt{2mK.E.}$

Therefore , de Broglie wavelength :

$\lambda = h / \sqrt{2mK.E.}$

We need to compute this with $K.E. = 1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$.

Mass of electron , $m = 9.109 \times 10^{-31} \text{ kg}$

Step-by-step :

1 . Compute $2mK.E. :$

$2 \times (9.109 \times 10^{-31}) \times (1.602 \times 10^{-19}) = 2.917 \times 10^{-49} \text{ kg} \cdot \text{J}$

$2 \cdot \sqrt{2mK.E.} = \sqrt{2.917 \times 10^{-49}} \approx 1.708 \times 10^{-24} \text{ kg} \cdot \text{m/s}$

3 . $h = 6.626 \times 10^{-34} \text{ J} \cdot \text{s}$

4 . $\lambda = 6.626 \times 10^{-34} / 1.708 \times 10^{-24} \approx 3.88 \times 10^{-10} \text{ m}$

Convert to cm : $3.88 \times 10^{-10} \text{ m} = 3.88 \times 10^{-8} \text{ cm}$

Wait ⚠ let's recheck units .

Actually , $1 \text{ m} = 100 \text{ cm}$ ⚡ so $3.88 \times 10^{-10} \text{ m} = 3.88 \times 10^{-8} \text{ cm}$

But look at the options : the choices are in cm , but one is given in m .

Let ⚠ convert to cm :

$3.88 \times 10^{-10} \text{ m} = 3.88 \times 10^{-8} \text{ cm}$

Wait ⚠ that's not matching any of the options .

Wait ⚠ $3.88 \times 10^{-10} \text{ m}$ is $3.88 \times 10^{-8} \text{ cm}$? Yes .

Check if option F is in meters : $5.27 \times 10^{-9} \text{ m}$ ⚡ is that close?

Wait , let ⚠ double-check the calculation .

Use standard formula for electron wavelength in meters :

$\lambda \text{ (in m)} = \sqrt{2 \times 9.11 \times 10^{-31} \times 1.602 \times 10^{-19}} / (6.626 \times 10^{-34})$

Wait better use standard shortcut :

$\lambda = h / \sqrt{2mE}$

Compute numerically :

$m = 9.11 \text{e-31 kg}$
 $E = 1.602 \text{e-19 J}$

So $p = \sqrt{2 \times 9.11 \text{e-31} \times 1.602 \text{e-19}} = \sqrt{2.917 \text{e-49}} = 5.401 \text{e-25 kg} \cdot \text{m/s}$

$h = 6.626 \text{e-34}$

$\lambda = 6.626 \text{e-34} / 5.401 \text{e-25} = 1.227 \text{e-9 m}$